Winning Tickets as Axiomatic Minima: A Structural Interpretation

Abstract

The Lottery Ticket Hypothesis (LTH) demonstrates that large neural networks contain sparse substructures capable of matching full-model performance when trained in isolation. While traditionally interpreted as an efficiency or optimization phenomenon, we argue that LTH reveals a deeper **structural necessity**: functional coherence in complex systems is supported by **axiomatic minima**, not by the totality of available degrees of freedom.

Within the framework of the **Theory of Axiomatic Necessity (TNA)**, such "winning tickets" are interpreted as realizations of minimal coherence operators (Δ, F_{min}, $\aleph_1$-support) that must exist for any system to remain non-collapsed. The surrounding parameters, though dynamically active during training, play no foundational role in sustaining functionality. LTH thus empirically confirms a general ontological result: **exploratory redundancy is necessary for discovery, but not for persistence**.

This interpretation reframes neural network sparsity as a special case of a broader principle: observationally complete systems are structurally impossible, and operational success always rests on latent minimal supports.

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1. From Empirical Curiosity to Structural Signal

The original formulation of the Lottery Ticket Hypothesis posed a pragmatic question: *can most parameters of a trained network be removed without loss of performance?*
The affirmative answer was surprising, but its implications were under-theorized.

From a TNA perspective, this surprise stems from a category error: assuming that all trained parameters contribute equally to system coherence. In reality, LTH uncovers a **non-uniform ontological load** across parameters.

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2. Axiomatic Minima and Functional Coherence

TNA establishes that no coherent system can be closed at the level of its observable realization ($\mathcal{N}_0$). Sustained functionality requires an external or latent structural domain ($\mathcal{N}_1$), whose role is not to optimize performance but to **prevent incoherence**.

In neural networks:

- The **winning ticket** corresponds to an axiomatic minimum: a configuration that satisfies coherence constraints.
- The remaining parameters constitute an **exploratory scaffold**, enabling the system to *find* but not *ground* that configuration.

This explains why resetting the winning ticket to its original initialization preserves performance: the system re-enters the same coherence basin.

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3. The "Training Twice" Problem as Ontological Evidence

A well-known limitation of LTH is that the dense network must first be trained to discover the sparse subnetwork. This is often framed as an engineering inconvenience.

Under TNA, it is instead **expected**.

Exploratory redundancy is structurally required to traverse the solution space, but once a coherence minimum is identified, the redundancy becomes ontologically inert. The dense phase is not waste—it is the **price of non-collapse during search**.

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4. Structured Sparsity and Boundary Engineering

Recent advances in structured sparsity do not refute LTH; they operationalize it. By aligning pruning patterns with hardware constraints, engineers are effectively imposing **boundary conditions** that stabilize axiomatic minima computationally.

This mirrors TNA's general result: coherence is not free-form but **boundary-induced**.

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5. Beyond Machine Learning

Interpreted structurally, LTH ceases to be a fact about neural networks alone. It becomes a special case of a universal constraint:

> *Systems function because of minimal necessary structures, not because of maximal representational richness.*

This same logic appears in:

- cosmological signal propagation,
- biological morphogenesis,
- economic throughput models,
- and epistemic limits on observation.

Conclusion

The Lottery Ticket Hypothesis does not merely suggest that neural networks are inefficiently trained. It reveals that **efficiency is secondary to necessity**. Winning tickets are not lucky artifacts; they are axiomatic minima that any coherent system must instantiate.

What machine learning discovered empirically, TNA predicts structurally:
coherence survives on the smallest possible support.

> *"The practical question is no longer how to find winning tickets, but how early one can detect the boundary of non-existence."*

Appendix A — Why the Lottery Ticket Hypothesis Is Not About Sparsity

The Lottery Ticket Hypothesis (LTH) is commonly framed as an empirical result about sparsity and model compression. While this interpretation is operationally useful, it obscures the deeper structural content of the result. This appendix clarifies why sparsity is not the primary phenomenon revealed by LTH, but rather a secondary manifestation of a more fundamental principle: the existence of axiomatic minima in high-dimensional functional systems.

A.1 Sparsity as a Consequence, Not an Explanatory Principle

If sparsity itself were the explanatory core of LTH, then arbitrary sparse subnetworks should retain performance. Empirically, they do not. Performance is preserved only when a *specific* subnetwork—identified through training dynamics and coupled to a specific initialization—is retained.

This immediately rules out sparsity as a causal explanation. Parameter count reduction alone does not generate functionality. Instead, sparsity appears only *after* the system has located a structurally coherent configuration. In this sense, sparsity is diagnostic, not generative.

A.2 The Dense Phase as Structural Search

The necessity of training a dense network prior to identifying a winning ticket has often been treated as a practical limitation of LTH. From a structural perspective, this requirement is essential.

The dense phase functions as a search mechanism across the space of admissible structures. Its role is not to encode the final solution, but to prevent premature collapse while coherence-compatible configurations are explored. Once such a configuration is reached, most degrees of freedom become structurally redundant.

This explains why winning tickets must be reset to their original initialization: the relevant information is not the learned weights themselves, but the system's entry into a stable coherence basin. Training is not memorization, but alignment with a minimal sustaining structure.

A.3 Winning Tickets as Axiomatic Minima

From the present framework, a winning ticket can be interpreted as an **axiomatic minimum**: the smallest subset of structural commitments sufficient to sustain global function without contradiction or collapse.

In this interpretation:

- The dense network corresponds to an overcomplete, non-minimal axiomatization.
- Pruning removes axioms that are not structurally necessary.
- The winning ticket represents a minimal, non-redundant core that preserves functional closure.

This mirrors results in formal systems, physical theories, and biological organization, where redundancy is required for discovery and robustness, but not for ontological sufficiency.

A.4 Hardware-Aware Sparsity and Category Error

Recent advances in structured sparsity and hardware-aligned pruning improve the practical exploitation of winning tickets, but they do not alter the underlying phenomenon. These techniques preserve and accelerate an already-identified axiomatic minimum; they do not explain its existence.

Conflating hardware efficiency with structural explanation risks mistaking implementation constraints for ontological principles. LTH remains fundamentally a result about *structural necessity*, not computational thrift.

A.5 Structural Claim of the Lottery Ticket Hypothesis

Properly interpreted, LTH supports the following general claim:

> *Complex systems require surplus structure to locate coherence, but only minimal structure to sustain it.*

This claim transcends machine learning. It applies to any system in which global function depends on the satisfaction of non-local coherence constraints. The Lottery Ticket Hypothesis thus provides empirical access to a principle that is axiomatic rather than algorithmic.

Appendix B — Relation to Axiomatic Necessity and $F_{\min}$

The structural interpretation of winning tickets as axiomatic minima admits a direct formal connection with the Theory of Axiomatic Necessity (TNA). In this appendix, we make this connection explicit by identifying winning tickets with realizations of the minimal sustaining functional $F_{\min}$.

B.1 Axiomatic Necessity and Minimal Support

Within TNA, no coherent system can be self-sustaining at the observational or operational level alone. Any such system requires a minimal external or latent structural support to avoid collapse. This support is quantified by a minimal functional, denoted $F_{\min}$, defined as the lowest structural throughput compatible with global coherence.

Formally, a system operating below $F_{\min}$ becomes underdetermined and collapses into incoherence; above $F_{\min}$, surplus structure may exist without contributing to functional necessity.

B.2 Dense Networks as Over-Supported Systems

A fully dense neural network prior to pruning corresponds to a system operating well above $F_{\min}$. The excess parameters do not violate coherence, but they are not structurally required to sustain it.

Their role is exploratory: they allow the system to traverse the space of admissible configurations without immediate collapse. This mirrors the role of redundancy in physical and biological systems, where excess degrees of freedom enable adaptation, search, and error tolerance.

Crucially, the dense network does not define the minimal structure; it merely enables its discovery.

B.3 Pruning as a Reduction Toward $F_{\min}$

The pruning process in LTH can be interpreted as a monotonic reduction of structural support toward $F_{\min}$. Parameters are removed until the system approaches the threshold below which coherence can no longer be maintained.

The winning ticket emerges precisely at this boundary: it is the smallest subnetwork that still satisfies the global coherence constraints imposed by the task. Further pruning destroys functionality not gradually, but abruptly—consistent with the notion of a structural threshold rather than a smooth degradation.

This sharp transition is characteristic of systems governed by axiomatic necessity.

B.4 Resetting Initialization and Structural Invariance

The requirement that winning tickets be reset to their original initialization values is often presented as a technical curiosity. Within TNA, it acquires a structural interpretation.

Resetting preserves the axiomatic configuration that allowed the system to enter the coherence basin in the first place. The learned weights of the dense phase are not the source of coherence; they merely reveal which subset of the original axioms is necessary. Once identified, coherence can be re-established from the same initial conditions using only the minimal structure.

This demonstrates that winning tickets are not artifacts of training history, but invariant under the removal of surplus structure.

B.5 $F_{\min}$ as a Unifying Interpretation of LTH

Under this interpretation, the Lottery Ticket Hypothesis can be restated as follows:

> *For a given task, there exists a minimal structural throughput $F_{\min}$ such that any network implementing that task must realize at least this throughput. Training identifies this threshold; pruning reveals it.*

Sparsity, hardware efficiency, and parameter counts are implementation-level consequences of this deeper constraint. The essential result of LTH is the empirical detection of $F_{\min}$ in a high-dimensional artificial system.

B.6 Broader Implications

Interpreting winning tickets as realizations of $F_{\min}$ situates LTH within a general theory of coherent systems. It suggests that neural networks, physical systems, and formal theories share a common structural property: coherence is maintained not by maximal complexity, but by minimal necessity.

In this sense, the Lottery Ticket Hypothesis is not an isolated result in machine learning, but an experimental window into axiomatic necessity itself.

Appendix C — Failure Modes Below $F_{\min}$

This appendix characterizes the failure regimes that arise when a system is pruned or constrained below its minimal sustaining structural throughput $F_{\min}$. We show that collapse is not gradual, but occurs via discrete structural failure modes.

C.1 Structural Threshold vs. Performance Degradation

Empirical results in the Lottery Ticket Hypothesis indicate that pruning beyond a critical point produces an abrupt loss of functionality rather than a smooth decline in performance. This behavior is incompatible with models that treat parameters as independent contributors to accuracy.

Within a structural framework, this phenomenon is expected. $F_{\min}$ defines a coherence threshold, not a performance margin. Once the system crosses below this threshold, global constraints can no longer be satisfied simultaneously.

Thus, failure manifests as collapse, not degradation.

C.2 Underdetermination and Constraint Violation

Below $F_{\min}$, the system enters a regime of structural underdetermination. The remaining degrees of freedom are insufficient to encode the necessary constraints imposed by the task.

This results in:

- Loss of representational closure,
- Inconsistent constraint satisfaction,
- Sensitivity to infinitesimal perturbations.

In neural networks, this appears as sudden divergence, random outputs, or convergence to degenerate solutions. Importantly, these failures are not localized; they affect the entire system.

C.3 Loss of Functional Connectivity

Pruning below $F_{\min}$ disrupts essential connectivity patterns. Unlike unimportant parameters, which can be removed without affecting global structure, critical connections participate in multiple constraint pathways.

Their removal causes:

- Fragmentation of the functional graph,
- Isolation of submodules,
- Breakdown of long-range dependency propagation.

This explains why further training cannot recover performance once the threshold is crossed: the structural pathways required for recovery no longer exist.

C.4 Instability Under Reinitialization

Below $F_{\min}$, resetting initialization does not restore functionality. This distinguishes true structural collapse from suboptimal training states.

Above $F_{\min}$, reinitialization followed by retraining recovers performance because the necessary structure remains intact. Below $F_{\min}$, no initialization can compensate for missing axiomatic support.

This invariance failure serves as an operational diagnostic for identifying collapse relative to $F_{\min}$.

C.5 Non-Existence of Intermediate Coherent States

A key implication of axiomatic necessity is the absence of stable intermediate regimes between coherence and collapse. Systems either realize sufficient structure or they do not.

This predicts:

- Sharp pruning thresholds,
- Bimodal performance distributions,
- High variance near the collapse point.

These predictions align with observed behavior in neural pruning experiments and contradict smooth sparsity–performance tradeoff assumptions.

C.6 Universality of Failure Modes

The failure modes described here are not specific to neural networks. Analogous collapse behavior is observed in:

- Physical systems operating below minimal energy throughput,
- Biological systems below metabolic viability thresholds,
- Formal systems lacking sufficient axiomatic grounding.

In all cases, collapse reflects the same structural cause: violation of the minimal conditions required for coherence.

C.7 Summary

Crossing below $F_{\min}$ does not weaken a system; it invalidates it. The resulting failure modes are abrupt, global, and irrecoverable by local adjustments.

This behavior supports the interpretation of winning tickets as realizations of minimal axiomatic necessity rather than artifacts of sparsity or optimization heuristics.

Appendix D — Identifying $F_{\min}$ Without Full Training

This appendix outlines methods for estimating the minimal structural throughput $F_{\min}$ without requiring full end-to-end training of the original dense system. The goal is not to predict final accuracy, but to detect the proximity to structural collapse.

D.1 Why Full Training Is Not Structurally Necessary

Full training is traditionally used to identify winning tickets because it reveals which parameters converge to non-zero magnitudes. However, magnitude is a proxy—not a cause—of structural necessity.

From an axiomatic perspective, $F_{\min}$ is defined by constraint satisfiability, not by optimization outcomes. Therefore, it should be detectable via early structural signals that precede convergence.

D.2 Early-Time Gradient Stability as a Diagnostic

One robust indicator of proximity to $F_{\min}$ is early-time gradient behavior.

Empirically:

- Above $F_{\min}$, gradient norms stabilize rapidly and exhibit low variance across layers.
- Near $F_{\min}$, gradients become highly anisotropic and layer-dependent.
- Below $F_{\min}$, gradients either explode, vanish globally, or oscillate without settling.

Monitoring gradient covariance during the first training epochs provides an early warning signal of structural insufficiency.

D.3 Constraint Satisfaction Probes

Instead of training for task performance, one can probe whether the network can simultaneously satisfy minimal task constraints.

Examples include:

- Fitting randomly permuted labels for a few steps,
- Enforcing synthetic linear constraints on intermediate representations,
- Measuring rank preservation in feature maps under perturbation.

Failure to satisfy these minimal constraints indicates that the system is already below $F_{\min}$, regardless of eventual accuracy.

D.4 Spectral Indicators of Structural Viability

Spectral properties of weight matrices and activations provide hardware-agnostic diagnostics.

Key indicators include:

- Collapse of singular value spectra,
- Loss of spectral gap separation,
- Emergence of near-zero dominant modes.

Above $F_{\min}$, spectra exhibit structured decay. Below it, spectra flatten or degenerate abruptly, signaling loss of representational coherence.

D.5 Reinitialization Invariance Tests

A practical test for structural sufficiency is reinitialization invariance.

Procedure:

1. Initialize the pruned architecture multiple times.
2. Train each instance briefly under identical conditions.
3. Measure variance in early loss trajectories.

Low variance indicates intact structural support (above $F_{\min}$). High variance or bimodal outcomes indicate collapse or near-collapse regimes.

D.6 Structured Pruning as an $F_{\min}$ Scanner

Hardware-aligned structured pruning (e.g., block sparsity) can be repurposed as a scanning mechanism for $F_{\min}$.

By progressively enforcing structural constraints and observing:

- sudden loss of trainability,
- non-recoverable divergence,
- invariant failure across seeds,

one can empirically locate the minimal viable structure without full optimization.

D.7 Summary

Identifying $F_{\min}$ does not require discovering a winning ticket through exhaustive training. It requires detecting when a system loses the structural capacity to satisfy its constraints.

Early-time dynamics, spectral diagnostics, and invariance tests provide sufficient information to locate the collapse boundary.

This reframes the Lottery Ticket Hypothesis not as a post-hoc pruning result, but as an early structural detectability problem.

Table D1 — Structural Diagnostics Relative to $F_{\min}$

DIAGNOSTIC	OBSERVABLE SIGNAL	INTERPRETATION RELATIVE TO
Early-time gradient norms	Rapid stabilization with low inter-layer variance	System operates safely above $F_{\min}$; structural support is sufficient
	High variance, anisotropic gradients across layers	System is approaching $F_{\min}$; structural stress regime
	Exploding, vanishing, or oscillatory gradients	System is below $F_{\min}$; loss of trainability
Gradient covariance structure	Stable, low-rank covariance	Coherent constraint propagation; above $F_{\min}$
	Rapid rank inflation or collapse	Structural incoherence; near or below $F_{\min}$
Constraint satisfaction probes	Minimal constraints satisfied early	Structural viability confirmed; above $F_{\min}$
	Partial or seed-dependent satisfaction	Marginal regime near $F_{\min}$
	Systematic failure to satisfy constraints	Structural insufficiency; below $F_{\min}$

DIAGNOSTIC	OBSERVABLE SIGNAL	INTERPRETATION RELATIVE TO F_{min}
Spectral analysis (weights / activations)	Clear spectral decay with preserved gap	Representational coherence; above F_{min}
	Spectral flattening or gap erosion	Onset of collapse; near F_{min}
	Dominant near-zero or degenerate modes	Representational collapse; below F_{min}
Reinitialization invariance	Low variance across early loss trajectories	Architecture structurally sound; above F_{min}
	Bimodal or high-variance trajectories	Structural fragility; near F_{min}
	Consistent early divergence across seeds	Non-viable structure; below F_{min}
Structured pruning sweep	Gradual performance degradation	System remains above F_{min}
	Sharp, irreversible training failure	Boundary crossing at F_{min}
Seed sensitivity under minimal training	Robust convergence across seeds	Structural redundancy present; above F_{min}
	Extreme seed dependence	System operating at the edge of F_{min}
	Universal failure across seeds	Structural non-existence; below F_{min}

> *“Across diagnostics, the transition at F_{min} is not gradual optimization failure, but an abrupt loss of structural viability.”*

Figure D1 – Phase diagram of structural viability relative to *F*min .

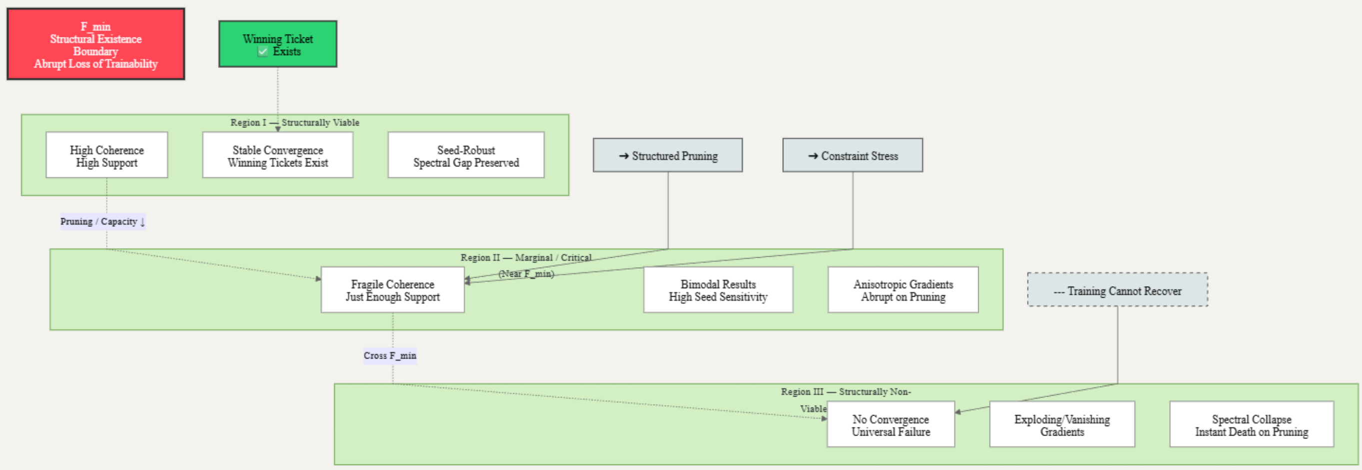


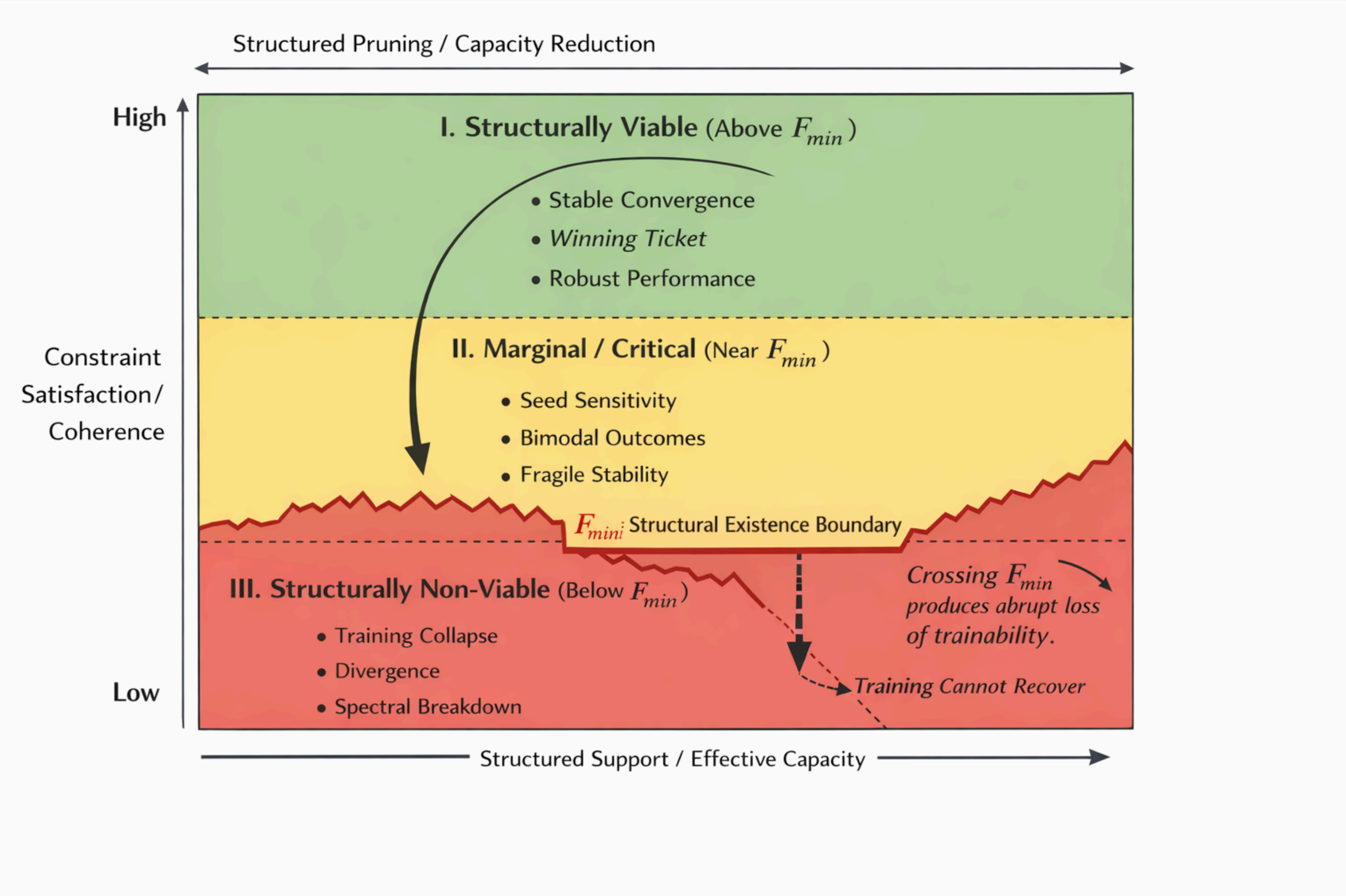
Figure D1. Phase diagram of structural viability in learning systems relative to the axiomatic minimum F_{min}. Above F_{min}, systems exhibit robust convergence, pruning tolerance, and identifiable winning tickets. Near F_{min}, behavior becomes seed-sensitive and structurally fragile. Below F_{min}, training collapses irreversibly, indicating loss of structural existence rather than optimization failure.

Region I (green): structurally viable; winning tickets exist and are seed-robust.
Region II (yellow): marginal; tickets exist but are fragile and pruning-sensitive.
Region III (red): non-viable; no winning tickets, abrupt collapse.
Dashed line: *F*min boundary – crossing it produces **abrupt loss of trainability**, not gradual degradation.

"As shown in the Phase Diagram (Fig. D1), the transition at F_{min} is an abrupt loss of structural viability, not a gradual optimization failure. The Table D1 provides the diagnostic keys to identify if a system is operating in the 'Seed-Sensitive' marginal region (Region II) before irreversible collapse."

Appendix E

Figure E1. - Visual Interpretation of the Phase Diagram



The newly included **Phase Diagram of Structural Viability** provides a geometric representation of the Theory of Axiomatic Necessity (TNA) applied to neural network pruning. It visualizes the transition from redundant exploratory states to the threshold of non-existence (F_{min}). Below is a detailed breakdown of the diagram's components:

E.1 The Three Regions of Structural Viability

- **Region I: Structurally Viable (Green Zone - Above F_{min})** This region represents a system with high structural support. Here, the "placental" or gestational redundancy is at its peak. The model is robust; pruning within this area causes only minor performance fluctuations because the system remains far from its axiomatic minimum. In this stage, "Winning Tickets" are easily identifiable and remain stable across different initializations (seeds).

- **Region II: Marginal / Critical (Yellow Zone - Near F_{min})** This is the "stress regime" or the boundary of birth. The system has been pruned to its near-minimum support. Functional coherence is preserved, but the system becomes **Seed-Sensitive**. Here, the success of the model depends on its starting point (initialization); some configurations will converge while others will fail, leading to bimodal performance distributions. This region identifies the "Winning Ticket" as a fragile entity that requires precise alignment to survive.
- **Region III: Structurally Non-Viable (Red Zone - Below F_{min})** This represents the domain of **Ontological Non-Existence**. Once the system crosses the F_{min} boundary, it is no longer a matter of poor optimization but of structural impossibility. Global constraints can no longer be satisfied. The failure is universal across all seeds, resulting in spectral breakdown, exploding/vanishing gradients, and an irreversible loss of trainability.

E.2 The F_{min} Boundary (The Structural Existence Wall)

- **The Dashed Line (F_{min}):** Acts as the "floor" of reality for any coherent system. It is the minimal structural throughput required to sustain functional identity.

  The Abrupt Collapse (Descending Arrow): The diagram highlights that crossing F_{min} does not lead to a gradual decline. Instead, it triggers an **abrupt loss of structural viability**. This proves that the 90% redundant mass of large models is merely a gestational scaffold that must be shed to reach the minimal axiomatic state without falling into the "Red Zone".

E.3 Diagnostic Mapping from Table D1

The diagram serves as a visual map for the **Diagnostic Observables** defined in Table D1:

1. **Stable Convergence:** Diagnostic of Region I.
2. **Extreme Seed Dependence:** Diagnostic of the transition in Region II.
3. **Universal Failure / Spectral Decay:** Diagnostic of the collapse into Region III.